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Received: 26 December 2025

Accepted: 18 June 2026

Cite this article as: Shen, A., Gong, C., Dong, Z. *et al.* Resonance-induced electron delocalization for efficient lead-free perovskite solar cells. *Commun Mater* (2026). <https://doi.org/10.1038/s43246-026-01252-4>

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Resonance-Induced Electron Delocalization for Efficient Lead-Free Perovskite Solar Cells

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Abstract: Tin perovskites have emerged as promising lead-free alternatives for eco-friendly perovskite solar cells; however, their device performance still lags behind that of lead-based counterparts, mainly due to poor thin-film quality, high defect densities, and the intrinsic instability Sn^{2+} . Here, we report a resonance-induced electron delocalization strategy to fundamentally stabilize tin perovskite by enhancing Sn-centered interactions using trithiocyanuric acid (Cy-SH) and cyanuric acid (Cy-OH) molecules as interfacial layer. We further demonstrate the electron delocalization between the $\text{N}-\text{C}=\text{O}$ and $\text{N}=\text{C}-\text{OH}$ tautomeric configurations of Cy-OH at the perovskite buried interface, which strengthens its interaction with uncoordinated Sn atoms, thereby significantly suppressing non-radiative recombination and enhancing carrier extraction and transport. As a result, the best-

performing device achieves a power conversion efficiency (PCE) of 12.1%, maintaining 86% of its initial efficiency after 1000 h aging in N₂ containing 50-100 ppm O₂. This work demonstrates that tautomeric molecules can be rationally designed and utilized to regulate tin perovskite crystallization and defect states, thereby providing guidelines for the development of high-performance lead-free photovoltaics.

Key words: Lead-free perovskite solar cells; Resonance-induced electron delocalization strategy; Tautomeric configurations, Power conversion efficiency , Stability

Introduction

Metal halide perovskites have attracted intense attention from both academia and industry as promising low-cost, next-generation thin-film photovoltaic materials.¹⁻³ However, their reliance on toxic lead remains a major obstacle to large-scale commercialization.⁴⁻⁶ To mitigate the environmental and health concerns, lead-free tin perovskites are regarded as the most promising alternatives due to their high absorption coefficient, excellent charge carrier mobility, low exciton binding energy, and environmental friendliness.^{7,8} Nevertheless, the performance of tin-based perovskite solar cells (TPSCs) still lags far behind their Pb-based counterparts primarily due to the rapid crystallization of tin perovskites and the susceptibility of Sn^{2+} to oxidation.⁹⁻¹¹ Most of the strategies developed for improving TPSCs have focused on suppressing the oxidation of Sn^{2+} typically through the incorporation of Sn compensators or reducing agents.^{12,13} However, these approaches are generally insufficient for effectively mitigating the defects that form within the resulting perovskite films. In addition, the residual chemical additives can disrupt the structural order of the tin perovskite lattice, leading to Anderson localization, which severely compromises the optoelectronic properties.¹⁴ So far, the power conversion efficiency (PCE) of TPSCs has exceeded 16%.⁸ This relatively low PCE reflects our still limited understanding of the intrinsic instability of Sn^{2+} . More in-depth investigations are therefore required to further improve both the efficiency and long-term stability of TPSCs.

The facile oxidation of Sn^{2+} fundamentally arises from its delocalized $5s^2$ lone pair electrons, which readily promote undesirable p-type self-doping.¹⁵⁻¹⁷ Ye et al. reported a localized-electron engineering strategy to improve the performance of TPSCs.¹⁸ This suggests that forming strong chemical coordination with Sn^{2+} can effectively suppress electron delocalization of Sn atom, thereby enhancing device performance. Recently, our group demonstrated that organic molecules with resonance structures capable of electron delocalization can enhance their interaction with uncoordinated Pb^{2+} and halide ionic vacancies defects during perovskite crystallization.^{1,3,19} Yu et al. also utilized conjugated triazine-based small molecules to fabricate high-performance Pb-based devices.²⁰ Such smart molecules are therefore expected to form much stronger coordination interactions with Sn^{2+} , enabling more effective control tin crystallization and defect passivation—both of which are crucial for achieving high-performance devices.^{21,22} Cyanuric acid is a typical electron delocalized molecule with a resonance-stabilized structure, featuring low electronegativity and strong electron-donating ability, favoring the interaction with Sn^{2+} .^{23,24} This can mitigate the adverse effects associated with Anderson localization of Sn atoms and promote a more ordered and stable crystal structure in tin perovskites.

Here, we report a resonance-induced electron delocalization strategy to fundamentally stabilize tin perovskite by using trithiocyanuric acid (Cy-SH) and cyanuric acid (Cy-OH) molecules as interfacial layer. We further demonstrated the

electron delocalization between the $\text{N}=\text{C}=\text{O}$ and $\text{N}=\text{C}-\text{OH}$ tautomeric configurations of the Cy-OH at the buried interface of tin perovskite. This tautomerization strategy strengthens the interaction with uncoordinated Sn atoms, thereby significantly suppressing non-radiative recombination and enhancing carrier extraction and transport efficiency. As result, the high-quality tin perovskite films were formed for the TPSCs. The resonance induced TPSCs based on Cy-OH achieved a PCE of 12.1% with enhanced stability, retaining 86% of its initial efficiency after 1000 h aging in N_2 . This work demonstrates that the molecules based on $\text{N}=\text{C}=\text{O}$ and $\text{N}=\text{C}=\text{S}$ resonance structures can massively control tin perovskite crystallization and defect regulation, thus providing important guidelines for developing high-performance lead-free perovskite photovoltaics.

Results and discussion

Research on the interaction between Cy-OH and perovskite

In this study, Cy-OH and Cy-SH tautomeric molecules with resonance structures ($\text{N}=\text{C}-\text{OH}$ or $\text{N}=\text{C}-\text{SH}$) were employed to regulate defect states and crystallization behavior in TPSCs, Cy-OH exhibits the optimal efficiency, and Cy-SH is subsequently used as a reference for discussion. To elucidate the influence of these molecules on device performance and operational stability, TPSCs were fabricated with the architecture of ITO/PEDOT:PSS/Cy-OH (or Cy-SH)/perovskite/ICBA/BCP/Al (Figure 1a). The tautomeric molecules were introduced at the buried interface between PEDOT:PSS and the tin perovskite layer, serving as an interfacial layer. These tautomeric molecules spontaneously self-assemble via intermolecular hydrogen bonding, forming a stable and dense supramolecular network that acts as a scaffold for perovskite films.^{14,25,26} Besides, the interaction between the resonance structures of the molecules and the Sn atom in perovskite induces a structural transformation from the enol form to the keto form based on electron delocalization. This tautomerization stabilizes the molecules in their thermodynamically favored keto configuration, which contains strong electron-donating groups ($\text{C}=\text{O}$ or $\text{C}=\text{S}$). These functional groups can effectively coordinate with undercoordinated Sn^{2+} sites at the perovskite surface, thereby reducing defect states at buried surface. Beyond direct coordination, the extended conjugated supramolecular network establishes a unique electronic environment that modulates

both the electronic structure and chemical stability of tin perovskites, even in regions not directly involved in chemical interactions with the molecules.¹⁴

To understand the chemical coordination interaction the Cy-OH and tin perovskite, we then performed Fourier-transform infrared spectroscopy (FTIR) measurements (Figure 1b and Figure S1 a).²⁷ After mixing SnI₂ with Cy-OH or Cy-SH, respectively, the stretching vibration of the -OH group shifted from 3210 cm⁻¹ to a lower wavenumber of 3206 cm⁻¹, while the -SH stretching vibration shifted from 2911 cm⁻¹ to 2908 cm⁻¹, confirming the coordination interactions between Sn²⁺ and the -OH/-SH groups. Furthermore, for the SnI₂/Cy-OH mixture, a new and distinct absorption peak emerged at 1698 cm⁻¹, which is attributed to the C=O stretching vibration. This observation indicates that Cy-OH underwent a structural transformation from the N=C-OH to N-C=O upon interaction with SnI₂, providing direct spectroscopic evidence for the tautomerization of Cy-OH induced by the Lewis acidic SnI₂. To further verify that the observed tautomerization is not a general phenomenon arising from any coordinating group, we conducted a control experiment using 5-amino-1-(2-hydroxyethyl)pyrazole (AHP), which contains similar coordinating groups (C=N and -OH) but lacks the ability to undergo resonance tautomerism. As shown in (Figure S1b), upon mixing with SnI₂, AHP exhibits no new C=O absorption peak, and no structural transformation is observed. This stark contrast provides strong evidence that the tautomerization behavior is unique to Cy-OH and plays a critical role in its superior interfacial modification

performance.

Subsequently, ^1H nuclear magnetic resonance (NMR) measurements (Figures S2-S3) clearly show significant shifts in the aromatic proton signals of both Cy-OH and Cy-SH upon the addition of SnI_2 . The chemical shifts of the resonance signals of Cy-OH and Cy-SH indicate the formation of hydrogen bonds in the form of $\text{N}=\text{C}-\text{OH}\cdots\text{I}^-$ and $\text{N}=\text{C}-\text{SH}\cdots\text{I}^-$. The broad resonance signal of $\text{SnI}_2\text{-CyOH}$ and $\text{SnI}_2\text{-CySH}$ suggests that the molecules are immobilized by the hydrogen bonds, which prevents the migration of I^- . These observable shifts provide direct evidence for a strong chemical interaction between the tautomeric molecules and the tin perovskite precursor.²⁸

To further elucidate the tautomerization behavior from an electronic structure perspective, X-ray photoelectron spectroscopy (XPS) measurements were carried out (Figure 1c).^{28,29} The O 1s spectrum of Cy-OH on the perovskite exhibited significant changes compared with that of the pristine Cy-OH sample. Specifically, the peak component associated with the keto form ($\text{N}-\text{C}=\text{O}$) shifted from 531.4 eV to 531.9 eV, indicating coordination with Sn^{2+} . Meanwhile, the relative area ratio of the $\text{N}-\text{C}=\text{O}$ component increases significantly from 57.7% to 75.6%, while the relative area ratio of the $\text{N}=\text{C}-\text{OH}$ component decreases from 42.3% to 24.4%. The pronounced variation in the relative proportion of these resonance-associated configurations indicates substantial interfacial electron redistribution after coordination between Cy-OH and the perovskite surface, providing direct

spectroscopic evidence for electron rearrangement at the molecular–perovskite interface.³⁰ In addition, single-crystal X-ray diffraction analysis reported in the literature further supports the existence of strong resonance characteristics in Cy-OH.³¹ The measured C=O bond length (1.214 Å) is longer than that of a typical C=O double bond (1.200 Å), while the C–N bond length (1.336 Å) is significantly shorter than a standard C–N single bond (1.445 Å). These abnormal bond-length characteristics indicate partial electron delocalization within the molecular backbone and confirm the coexistence of the N–C=O and N=C–OH resonance forms.

Meanwhile, after treatment with these resonance molecules, the Sn 3d and I 3d peaks of the perovskite lattice shifted toward lower binding energies, indicating an increased local electron density arising from electron donation and interfacial coordination due to the electron-rich carbonyl/thiocarbonyl (N–C=O/N–C=S) groups (Figure 1d, e).^{32,33} Moreover, the fitting of Sn 3d_{5/2} peaks revealed a significantly higher Sn²⁺/Sn⁴⁺ ratio in the treated film compared to the control (Figure S4). Together, these results reveal that the tautomeric molecules stabilize the perovskite surface via coordination while their electron-donating nature effectively inhibits Sn²⁺ oxidation, leading to substantially improved intrinsic oxidation resistance.^{34,35}

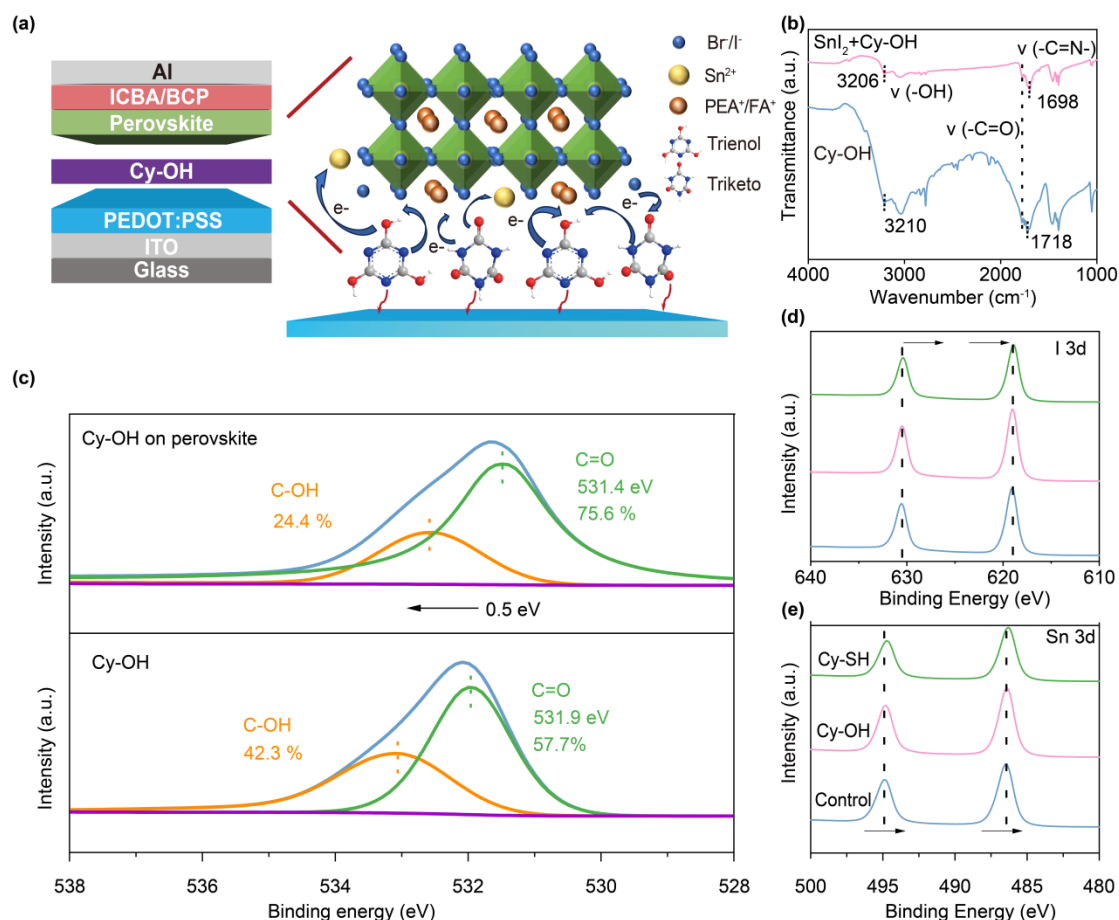


Figure 1. (a) Schematic illustration of the device architecture and the interaction mechanism between tautomeric molecules and tin perovskites in TPSCs. (b) FTIR spectra of Cy-OH and Cy-OH-SnI₂ samples. (c) XPS of Cy-OH film and Cy-OH deposited on the tin perovskite. (d, e) XPS measurements of (d) I 3d and (e) Sn 3d.

Research on the properties of perovskite films

To further investigate the influence of these tautomeric molecules on the structural and optical properties of tin perovskite films, a series of thin-film characterizations was carried out. First, from the X-ray diffraction (XRD) patterns (Figure 2a), all tin perovskite films exhibit characteristic diffraction peaks at approximately 14° and 28°, which can be indexed to the (100) and (200) crystal planes of three-dimensional (3D) tin perovskites, respectively.^{36,37} Notably, tin

perovskite films treated with the tautomeric molecules exhibited a pronounced enhancement in diffraction peak intensities. Among them, the Cy-OH-treated film showed the most prominent improvement, with the intensity of the (100) diffraction peak approximately doubling compared to that of the control sample. These results clearly demonstrate that the introduction of tautomeric molecules effectively promotes preferential crystal growth for high-quality tin perovskite film. The ultraviolet-visible (UV-vis) absorption spectra (Figure 2b) show that the target perovskite films have higher light absorption than the control sample, indicating enhanced light-harvesting capability for the devices.^{38,39} These enhanced light-absorption capability should be due to the improved crystalline quality and the reduced defect density at buried interface.⁴⁰ As depicted in the Tauc plot derived from the UV-visible spectrum (Figure 2c), the band gaps of all tin perovskite films were determined to be approximately 1.37 eV, indicating that the introduction of the tautomeric molecule does not alter the intrinsic band structure of the tin perovskite. Additionally, Ultraviolet Photoelectron Spectroscopy (UPS) (Figure S12) measurements reveal that the treated device exhibits a smaller energy level barrier, which is relatively more favorable for hole transport.³⁵ This key finding suggests that the Cy-OH and Cy-SH molecules primarily function at the perovskite surfaces and/or grain boundaries, rather than being incorporated into the crystal lattice as dopants.⁴¹ In addition, atomic force microscopy (AFM) revealed that the Cy-OH-treated perovskite film exhibits a smoother surface (the RMS roughness reduced

from 20.02 nm to 11.2 nm), further confirming the positive role of tautomeric molecules in optimizing film quality (Figure S5). To investigate the real-time crystallization process of tin-based perovskite films during annealing, in-situ photoluminescence (Figure S6) and in-situ UV-vis absorption spectra (Figure S7) were employed. The Cy-OH-based film showed much higher PL intensity than the control, indicating improved crystalline quality. From the first 100 s before annealing (Figure S6c, d), the control film's PL intensity began to rise at around 30 s, marking the onset of recrystallization, whereas the Cy-OH-treated film showed signal emergence earlier (25 s), indicating accelerated nucleation. Furthermore, in-situ UV-vis spectroscopy revealed that the Cy-OH-containing film exhibited a slow increase in absorption intensity after 10 s, while the control film showed a rapid increase within 5 s, confirming that Cy-OH significantly retards the crystallization growth rate of tin-based perovskites.

Steady-state photoluminescence (PL) and time-resolved photoluminescence (TRPL) measurements were also performed on the corresponding perovskite films (Figure 2d, e). The steady-state PL spectra (Figure 2d) reveal a pronounced enhancement in PL intensity for the Cy-OH and Cy-SH treated films compared with the control film. For TRPL measurement of the corresponding tin perovskite films with a hole-free structure (Figure 2e), the average carrier lifetime (τ_{avg}) was extended to 19.17 ns and 28.00 ns for the Cy-SH- and Cy-OH-treated films, respectively, which are significantly longer than that of the control film (15.13 ns), indicating

reduced the defects and suppressed non-radiative recombination after the introduction of the tautomeric molecules.^{19,42} Moreover, in the tin perovskite films base on a hole transport layer (Figure 2f), the carrier lifetime exhibits a distinctly different trend. The treated films show a faster PL quenching behavior, with average lifetimes of 11.79 ns and 9.17 ns for the Cy-SH and Cy-OH treated samples, respectively, which are significantly shorter than that of the control film (14.52 ns). This accelerated PL quenching is not associated with defect-assisted recombination but is instead attributed to more efficient carrier extraction and injection at the perovskite/HTL interface.⁴³

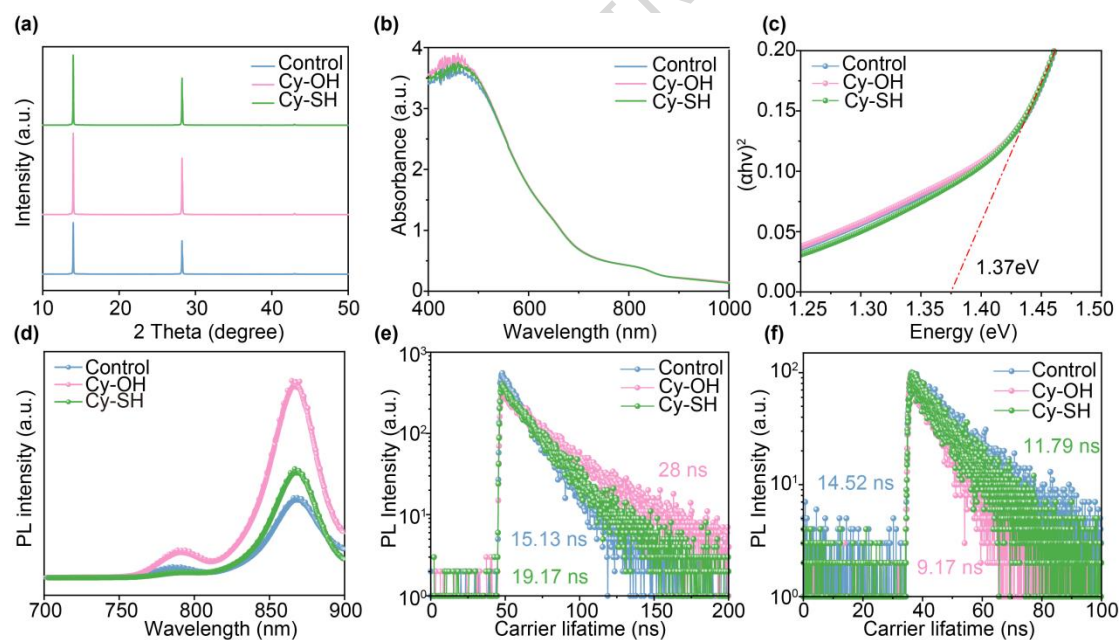


Figure 2. (a) XRD patterns of perovskite films with and without tautomeric molecules treatments. (b) UV-vis absorption spectra. (c) Tauc plots of the control, Cy-OH, and Cy-SH-treated perovskite films. (d) PL of the corresponding perovskite films. (e, f) TRPL decay curves of the corresponding perovskite films (e) without and (f) with a hole transport layer.

Research on photovoltaic performance and device stability

Encouraged by the above benefits related to the tautomeric molecules, we fabricated inverted TPSCs for investigate device performance. To determine the optimal concentration for buried interface modification, as well as to rule out the influence of molecules that lack resonance tautomerism but possess similar coordinating groups we conducted a series of concentration-dependent experiments (Table S1 S2). Among the tested concentrations, the device modified with Cy-OH at 2 mg/mL exhibited the highest power conversion efficiency (PCE). The current density-voltage (J - V) curves of the corresponding devices are shown in Figure 3a. The control device achieved a PCE of 9.9%, with a short-circuit current density (J_{SC}) of 17.10 mA cm⁻², an open-circuit voltage (V_{OC}) of 0.80 V, and a fill factor (FF) of 72.05%. In contrast, the target treated with the Cy-OH exhibited a superior PCE of 12.1%, along with a J_{SC} of 19.29 mA cm⁻², an enhanced V_{OC} of 0.86 V, and an FF of 72.80%. From the forward and reverse scan J - V curves (Figure 3b, c) the target device based on Cy-OH show a hysteresis index (HI) of 0.06, significantly lower than that of the control device (HI=0.10). To evaluate reproducibility, 15 independent devices were fabricated and measured (Figure 3d and Figure S8). The Cy-OH-treated devices show a higher average performance compared to the control group with good reproducibility. The external quantum efficiency (EQE) spectrum (Figure 3e) demonstrates that the target device exhibits higher quantum efficiency across the spectral range, and integrated J_{SC} values of the devices is in good agreement with the J - V measurement results.

We then investigated the long-term stability of the corresponding devices under various conditions (Figure 3f and Figure S9). The device based on Cy-OH molecule exhibited excellent storage stability, retaining 86% of its initial PCE after 1000 h under N₂ containing 100–200 ppm of O₂. In contrast, the control device degraded rapidly, retaining only 61% of its initial PCE after 1000 h of aging. Moreover, the photothermal stability of unencapsulated devices was investigated under continuous 1-sun illumination in an open-circuit configuration at $60 \pm 5^\circ\text{C}$. The target device maintained 60% of its initial efficiency after 250 hours, while the control device retained only 30%. Meanwhile, day–night cycling stability tests were conducted in N₂ containing 100–200 ppm O₂ with a 12-hour light–dark alternation. After 30 consecutive cycles (approximately 700 hours) in a nitrogen environment, the target device retained 85% of its initial efficiency, whereas the control device dropped significantly to 66%. To evaluate the effect of Cy-OH on the ambient stability of SnI₂, the color evolution of pristine SnI₂ and SnI₂ with Cy-OH was monitored upon air exposure for different time (Figure S10). The pristine SnI₂ film rapidly changed from yellow to dark brown within 15 min, indicating severe oxidation of Sn²⁺ to Sn⁴⁺. In contrast, the Cy-OH-modified SnI₂ film showed significantly retarded darkening, with only a slight change even after 25 min of air exposure. These results demonstrate that Cy-OH effectively suppresses oxidation and enhances the stability of SnI₂ under ambient conditions.

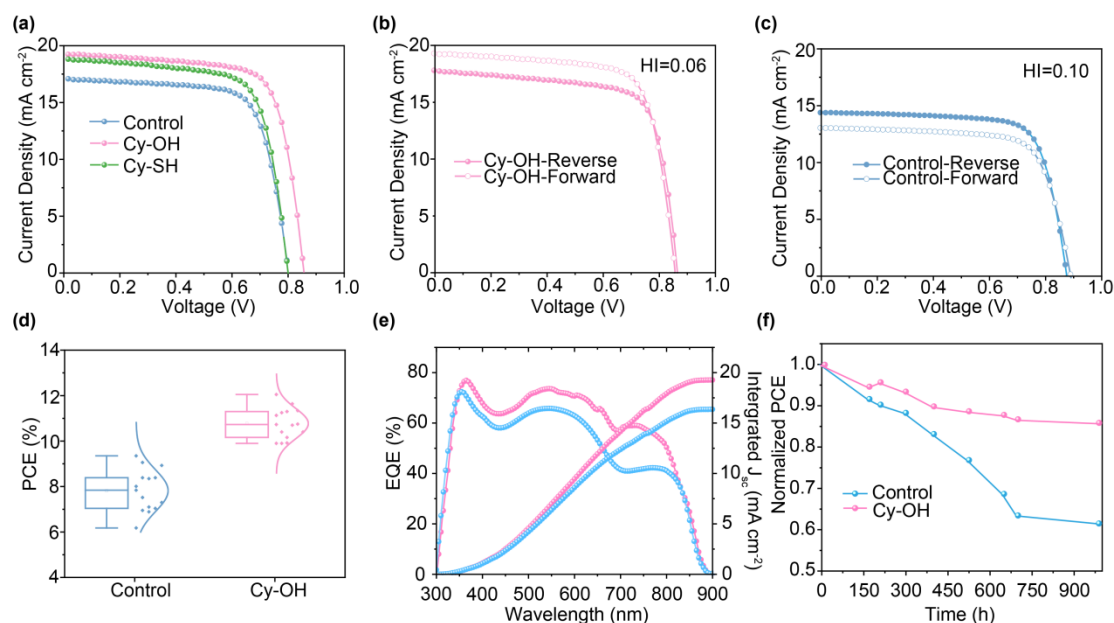


Figure 3. (a) $J-V$ curves of the control and target devices based on Cy-OH and Cy-SH molecules. (b, c) $J-V$ curves of the corresponding devices under reverse and forward scans. (d) The PCE Statistics for the devices. (e) EQE spectra and the integrated J_{sc} . (f) Storage stability of the unencapsulated devices in N_2 containing 50–100 ppm of O_2 at $25^\circ C$.

Research on carrier transport

Defect-induced non-radiative recombination and charge transport losses constitute a critical bottleneck in improving the performance of TPSCs. To gain deeper insight into the mechanism of the tautomeric molecules, we have provided the dark current density-voltage ($J-V$) curves. The devices treated with Cy-OH and Cy-SH exhibited a markedly reduced dark current density in the reverse-bias region compared with the control (Figure 4a), indicating these molecules suppressed non-radiative recombination and reduced the defects.^{44,45}

Furthermore, we performed light intensity-dependent V_{OC} measurements to evaluate non-radiative recombination in both the control and treated perovskite

devices. Here, $\eta kT/q$ represents the slope of the best-fit line obtained from the data. The ideality factors of the Cy-SH- and Cy-OH-treated perovskites were 1.42 and 1.43, respectively, significantly lower than those of the control devices (Figure 4b). The reduced η values further confirm a decrease in defect density and non-radiative recombination.

Furthermore, to quantitatively evaluate the defect density of devices incorporating the tautomeric molecules, space-charge limited current (SCLC) measurements were carried out using hole-only devices (Figure 4c). The trap-filling limit voltage (V_{TFL}) values for the devices treated with Cy-OH and Cy-SH reduced to 0.92 V and 0.94 V, respectively, significantly lower than that of the control device (1.16 V). The value of n_{trap} in the perovskite film is calculated as follows:

$$n_{trap} = \frac{2\varepsilon\varepsilon_0 V_{TFL}}{qL^2} \quad (1)$$

where ε_0 denotes the vacuum dielectric constant, ε represents the relative dielectric constant, q is the elementary charge, L is the thickness of the perovskite film, and V_{TFL} refers to the trap filled limit voltage.⁴⁶⁻⁴⁸ Thus the corresponding n_{trap} values are calculated as $1.86 \times 10^{16} \text{ cm}^{-3}$ and $1.90 \times 10^{16} \text{ cm}^{-3}$ for the Cy-OH and Cy-SH treated devices, respectively, much lower than that of the control sample ($2.36 \times 10^{16} \text{ cm}^{-3}$). This result further confirmed that introducing the tautomeric molecules effectively reduces the trap state density in the films and significantly suppresses trap-assisted carrier recombination.

Under the dark conditions, electrochemical impedance spectroscopy (EIS) was conducted to investigate the influence of the tautomeric molecules on the internal charge transport and recombination (Figure 4d). The Nyquist graph and equivalent circuit model for the comparison and improvement devices are displayed in Figure 4d.⁴⁹ The devices treated with Cy-OH and Cy-SH exhibited a significantly larger semicircle radius in the mid-to-low frequency region, indicating pronounced differences in their charge transport processes. The recombination resistance (R_{rec}) for the Cy-OH- and Cy-SH-based devices was approximately 1.5-fold and 1.3-fold higher, respectively, than that of the control device. These results indicate the suppression of charge recombination at the surface interface of the perovskite film due to the introduction of the tautomeric molecules.

To further study the influence of N-AE on the built-in potential (V_{bi}), the Mott-Schottky curves were conducted (Figure S11). The V_{bi} of the control device was 0.61 V, increasing to 0.65 V and 0.69 V for the Cy-OH and Cy-SH based devices, respectively. The observed enhancement in the built-in potential correlates well with the increased open-circuit voltage obtained from the J - V measurements.

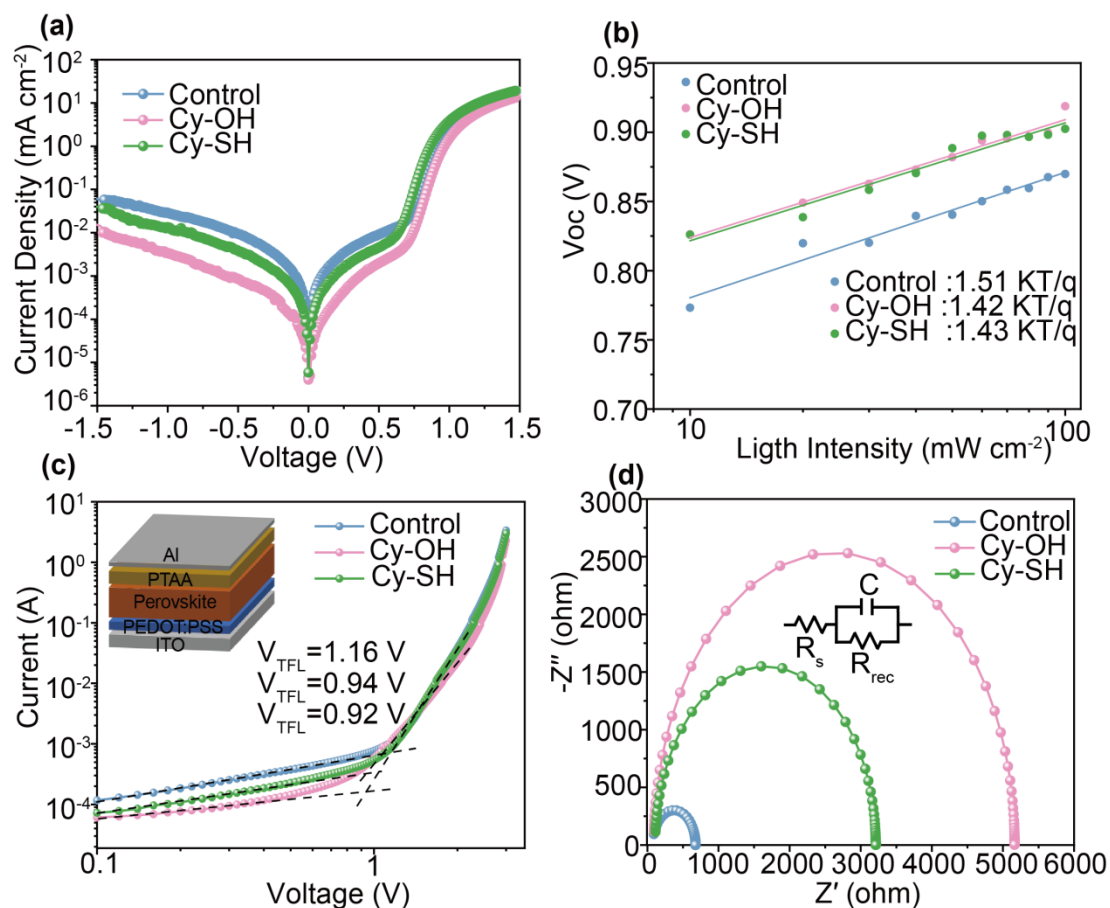


Figure 4. (a) $J-V$ curves under dark conditions. (b) V_{oc} -light intensity trend plots. (c) SCLC measurements of the hole-only devices. (d) EIS curves of the corresponding devices.

Conclusion

In summary, we developed a resonance-induced electron delocalization strategy to fundamentally stabilize tin perovskite for high-performance TPSCs. Cy-SH and Cy-OH molecules were employed as interfacial layers, where their tautomeric configurations enable enhanced interactions with Sn atoms. We also demonstrated the electron delocalization between the N-C=O and N=C-OH tautomeric configurations of the Cy-OH at the buried tin perovskite interface. This

resonance-induced electron delocalization strategy enables strong coordination with Sn^{2+} ions and modulates the electronic environment at the perovskite surface, thereby suppressing non-radiative recombination, regulating perovskite crystallization, and enhancing carrier extraction and transport. As a result, the optimized device achieves a champion PCE of 12.1%, significantly higher than that of the control device (9.9%), along with mitigated hysteresis. Besides, the device also exhibited outstanding long-term stability, retaining 86% of its efficiency after 1000 hours aging in N_2 .

Methods

Materials

Poly(3,4-ethylenedioxythiophene): Poly(styrenesulfonate) (PEDOT:PSS) solution was purchased from Heraeus (Clevios PVP. Al 4083). Formamidinium iodide (FAI) ($\geq 99.5\%$) and Phenethylammonium bromide (PEABr) ($\geq 99.5\%$) were obtained from Xi'an Polymer Light Technology Co., Ltd. Tin fluoride (SnF_2) (Sigma-Aldrich, 99%), ICBA (1-materia, $>99\%$), Bathocuproine (BCP, 99.7%) were purchased from Luminescence Technology Co., Ltd. N,N-dimethylformamide (DMF, 99.5%), dimethyl sulfoxide (DMSO, $> 99.9\%$), and chlorobenzene (CB, 99.5%) were from either Acros Organics, Alfa Aesar, or TCI (Shanghai) Development Co., Ltd. Al were purchased from Zhongnuo Advanced Material (Beijing) Technology Co., Ltd. Cyanuric acid (Cy-OH, 98%) and thiocyanuric acid (Cy-SH, 95%) were purchased from Shanghai Macklin Biochemical Co., Ltd. Toluene ($>99.5\%$,

Sinopharm 10022818) removed H₂O using a type 4A molecular sieve.

SnI₂ synthesis

10 mmol I₂ was dissolved in 2.5 ml DMSO completely, and then diluted by 10 ml DMF to form 0.8 M I₂ solution (DMSO: DMF = 1:4, v/v). The excessive tin powder was added into 0.8 M I₂ solution, and the solution was stirred at room temperature to yield SnI₂ solution in a glovebox.

Perovskite precursor

A 0.8 M perovskite precursor comprised PEABr, FAI, SnI₂ solution, and SnF₂ in the molar ratio of 0.15: 0.85: 1: 0.075 were mixed and stirred at 70°C for 1 hour.

Device fabrication.

The indium tin oxide (ITO) electrode on a glass substrate was initially cleaned with deionized water, followed by acetone and ethanol, each for 15 minutes using an ultrasonic cleaner. After being annealed at 100 °C for 15 minutes, the ITO surface underwent treatment with UV ozone for 15 minutes. Next, the PEDOT: PSS solution was spin-coated onto the ITO surface at 4000 rpm for 45 seconds, followed by annealing at 120 °C for 20 minutes. The precursor was then spin-coated in two steps: first at 1000 rpm for 10 seconds and then at 5000 rpm for 30 seconds. During the second stage, 600 ul of toluene was added to the surface at the 22-second mark, after which the substrate was annealed at 80 °C for 10 minutes. For the ICBA layer, the solution (20 mg/mL in chlorobenzene) was spin-coated at 1000 rpm for 30 s and subsequently annealed at 70 °C for 10 min. A saturated BCP solution in isopropanol was then spin-coated at 6000 rpm for 30 s, with the resulting film thermally annealed

at 70 °C for 10 minutes. Finally, aluminum (100 nm) was deposited.

Characterization

The fabricated TPSCs were measured without encapsulation at 25 °C in the glove box filled with N₂. Their photocurrent density-voltage (J-V) curves and power conversion efficiencies (PCEs) were obtained by a computer-programmed Keithley 2400 source/meter under 100 mW cm⁻² illumination under AM 1.5 G solar simulator (SAN-EI Electric Co., Ltd.). The external quantum efficiency (EQE) measurements were conducted in air by QE-R system (Enli Technology Co., Ltd.). The electrochemical impedance spectra (EIS) were measured on a CHI660 electrochemical workstation (CH Instrument Inc.). A 20 mV voltage perturbation was applied at different direct current voltages ranging from 0 to 0.8 V with frequencies between 1000000 and 1 Hz under dark conditions. The hole-only devices with the structure of ITO/PEDOT: PSS (~ 40 nm)/perovskite (~200 nm)/PTAA (10 nm)/Al (100 nm) were measured from 0 to 3 V with a step size of 0.02 V under dark conditions by a Keithley 2400 source/meter unit. UV-vis absorption spectra were obtained on a Jasco V-750 UV-Visible/Near-Infrared Spectrophotometers. Fourier-transform infrared spectroscopy was conducted utilizing the potassium bromide pellet technique for sample analysis. X-ray diffraction (XRD) was conducted on Rigaku Smart lab X-Ray diffractometer. X-ray photoelectron spectroscopy (XPS) experiments were carried out on an ESCALAB 250 system equipped with a monochromatic Al K α X-ray source (h ν =1486.6 eV).

Steady-state photoluminescence (PL) spectra were collected using F-4700 fluorospectrophotometer. ^1H -nuclear magnetic resonance (NMR) spectra were recorded on Bruker Ultra Shield Plus 400 MHz instruments with DMSO- d_6 as the solvents and tetramethylsilane (TMS) as the internal standard. The time-resolved PL decays of the perovskite films (TRPL) were measured by an Edinburgh FLS 980. A 660 nm pulse laser with a repetition frequency of $\sim 1\text{--}20$ MHz was employed.

Data availability

The authors declare that the data supporting the findings of this study is available within the article and its supplementary Information file. All other data that support the results of this study are available from the corresponding author upon reasonable request.

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Author Contributions

A.S. and C.G. jointly conceived the study. C.G., A.S., M.C., and Z.D. were responsible for conducting experiments, testing and analyzing data, and creating figures and tables. L.X., R.C., supervised the project. A. S. drafted the manuscript. A.S., C.G., Z.D. M.C., W.L., R.C., and L.X. participated in discussions regarding the results and contributed to the preparation and revision of the manuscript

Funding declaration

National Natural Science Foundation of China (62274094); State Key Laboratory of Silicon and Advanced Semiconductor Materials, Zhejiang University (SKL2024-04).

Acknowledgements

This study was supported by Huali Talents Program of Nanjing University of Posts and Telecommunications.

Competing interests

The authors declare no competing interests